ALWARDAS POLYTECHNIC

PROJECT TITLE:

Digital Forensic Investigation Toolkit — ForensiQ

with Automated Evidence Acquisition, Multi-Source Analysis, Artifact Correlation & Court-Ready Reporting

1. PROBLEM STATEMENT

Digital forensic investigations, whether conducted by law enforcement, corporate security teams, or academic labs, involve handling disk images, memory dumps, and network captures that are large, heterogeneous, and highly sensitive to mishandling. Manual acquisition, hashing, artifact extraction, and timeline reconstruction are time-consuming, error-prone, and difficult to standardize across investigators, and any break in the chain of custody can render evidence inadmissible.

This project introduces ForensiQ, an integrated digital forensic toolkit that automates write-blocked evidence acquisition, cryptographic integrity verification, disk/memory/network artifact analysis, cross-source timeline correlation, and generation of structured, court-ready investigation reports.

2. INNOVATION / NOVELTY

The core innovation lies in the seamless integration of multiple forensic sub-disciplines into a single, end-to-end investigation workflow:

- Write-blocked evidence acquisition with automatic hash verification
- Tamper-evident, auto-generated chain-of-custody logging
- Unified disk, memory, and network artifact analysis engine
- Cross-source timeline reconstruction from filesystem, registry, and log data
- Automated IOC and keyword correlation across all extracted artifacts
- Structured, legally-formatted PDF report generation

3. ISSUES ADDRESSED

- Manual, error-prone evidence acquisition and hashing
- Fragmented toolchains across acquisition, analysis, and reporting
- Inconsistent or missing chain-of-custody documentation
- Time-consuming manual timeline reconstruction
- Difficulty correlating artifacts across disk, memory, and network sources
- Non-standardized reports that weaken legal admissibility

4. OBJECTIVES

- Acquire disk, memory, and network evidence through write-blocked, hash-verified channels
- Auto-generate a tamper-evident chain-of-custody log for every case
- Parse file systems and recover deleted or fragmented files via file carving
- Analyze RAM dumps for processes, connections, and injected artifacts
- Reconstruct a unified timeline across all evidence sources
- Correlate artifacts against IOC and keyword watchlists automatically
- Generate structured, court-ready forensic reports

5. AIM OF THE PROJECT

To develop a unified digital forensic toolkit that automates evidence acquisition, multi-source artifact analysis, and correlation, and produces structured, chain-of-custody-backed reports to support faster, more reliable, and legally admissible investigations.

6. METHODOLOGY

Step	Phase	Description
Step 1	**Evidence Acquisition & Hashing**	Acquire disk, memory, and network images through write-blocked interfaces and compute SHA-256/SHA-1 hashes to verify integrity at capture.
Step 2	**Chain-of-Custody Logging**	Automatically record acquisition metadata, timestamps, and custodian actions into a tamper-evident, auditable log.
Step 3	**Filesystem & Artifact Parsing**	Parse NTFS/EXT/FAT structures and recover deleted or fragmented files using file-carving techniques.
Step 4	**Memory Forensics**	Analyze RAM dumps for running processes, network connections, and injected code using a Volatility-based pipeline.
Step 5	**Network Traffic Analysis**	Parse PCAP captures to extract sessions, protocols, and suspicious communication patterns.
Step 6	**Timeline Reconstruction**	Correlate filesystem, registry, and log timestamps into a single unified investigative timeline.
Step 7	**Report Generation**	Compile verified evidence, correlated artifacts, and chain-of-custody records into a structured, court-ready PDF report.

7. TECHNOLOGY STACK & IMPLEMENTATION

Layer	Technology	Purpose
Frontend	Flutter (Dart)	Cross-platform investigator dashboard & case UI
Backend	Rust — Axum	REST API, evidence processing & workflow orchestration
Disk Forensics	The Sleuth Kit / libtsk	Filesystem parsing, deleted-file recovery & carving
Integrity	SHA-256 / SHA-1	Cryptographic hashing for evidence verification
Database	PostgreSQL	Case data, evidence metadata & chain-of-custody logs
Auth & Storage	JWT + S3-compatible	Secure investigator authentication & evidence storage

8. SYSTEM WORKFLOW

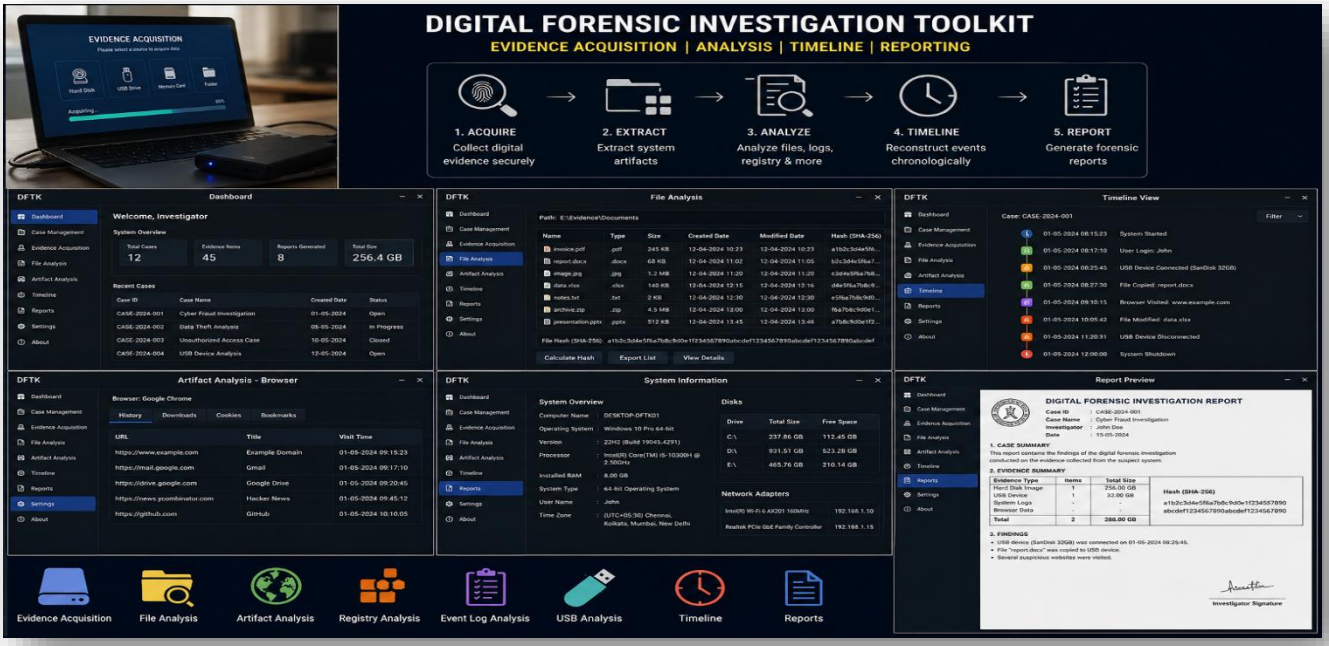
01	**Evidence Acquisition**
02	**Hashing & Chain of Custody**
03	**Filesystem & Memory Parsing**
04	**Network Traffic Analysis**
05	**Timeline Reconstruction**
06	**IOC & Artifact Correlation**
07	**Final Forensic Report**

9. RESULTS & EXPECTED BENEFITS

Faster Investigations	Automated acquisition-to-report pipeline significantly cuts manual analysis time.
Evidence Integrity	Cryptographic hashing and logging ensure tamper-evident evidence handling.
Unified Analysis	Single platform for disk, memory, and network artifacts instead of disjointed tools.
Legal Admissibility	Structured, chain-of-custody-backed reports support courtroom use.
Reduced Workload	Automates repetitive extraction, correlation, and documentation tasks.

10. EXPECTED OUTCOMES

- Fully automated evidence acquisition and hash-verification pipeline
- Tamper-evident, auto-generated chain-of-custody logging system
- Integrated disk, memory & network artifact analysis engine
- Cross-source timeline reconstruction with IOC correlation
- Structured, court-ready PDF forensic reports
- Investigator dashboard with case management and analytics
- Significant improvement in investigation turnaround time



PROJECT DETAILS:

COLLEGE	ALWARDAS POLYTECHNIC
DEPARTMENT	COMPUTER ENGINEERING
SUBJECT CODE	CM-509
SUBJECT	PROJECT WORK

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